NO.64 熱湯の方がより速く氷になる?(2)

Convinced that he had discovered something, Mpemba told a visiting physics professor about his accidental experiment. Like his teacher, the professor was a little doubtful, but invited him to test the theory.

The two began by filling 100 mL beakers with 70 ml samples of water of varying temperatures and placing them in the ice box of a normal refrigerator. What they noticed was that it took longer for the water to freeze when the temperature was at 25° C than when it was at a much hotter 90°C. Since then, the phenomenon has been known as the Mpemba effect. However, while the two were able to demonstrate it, neither could find a scientific explanation for why it occurred. Over the years, researchers have come up with several theories.

The theory that most believe is fairly straightforward. It is a known fact that hot water *evaporates faster than cold. Hence, when boiling water is tossed into cold air, some of it turns into steam and disappears, leaving behind less to turn to ice! Sounds plausible, right? In fact, Mpemba had thought of this possibility and even tested it. Unfortunately, he found no difference in the volumes of the ice formed at different temperatures.

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*　evaporate「蒸発する」

1	convince	他動詞	(人)を確信(納得)させる
2	accidental	形容詞	偶然の
3	traffic	名詞	交通(量)
4	disaster	名詞	(大)災害
5	insurance	名詞	保険
6	theory	名詞	理論
7	academy	名詞	(大学などの)高等教育機関・専門学校
8	fill	他動詞	を満たす・を占める
9	occupy	他動詞	を占める・を占領する
10	normal	形容詞・名詞	標準の・正常な・標準
11	standard	形容詞・名詞	標準の・標準・基準
12	refrigerator	名詞	冷蔵庫
13	notice	他動詞・名詞	に気づく・注意・通知
14	overlook	他動詞	を見落とす・を大目に見る・を見下ろす
15	neither	代名詞・副詞・形容詞	何方の～も…しない(人)・もまた…ない
16	nor	接続詞	S もまた…ない
17	occur	自動詞	起こる・(考えなどが突然)浮かぶ[to]
18	incident	名詞	出来事・事件
19	strike	他動詞・名詞	の心の中に浮かぶ・にぶつかる・打撃
20	several	形容詞	幾つかの・それぞれの・いろいろな
21	multiple	形容詞・名詞	多様な・倍数の・倍数
22	fairly	副詞	かなり・適切に・公平に
23	steam	名詞	上記
24	disappear	自動詞	消える
25	fade	自動詞	(色などが)褪せる・衰える
26	vanish	自動詞	(突然)消える
27	volume	名詞	量・容積・音量・1 冊

1	convince	
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26	vanish	
27	volume	